2 A roller coaster car of mass 380 kg slides down a smooth slope and executes a loop of diameter d. After the loop, it experiences friction only in the area denoted as zone A, as shown in Fig. 2.1.

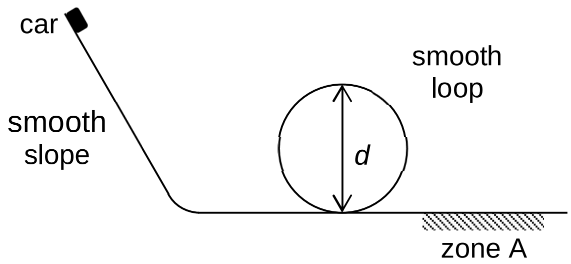

Fig. 2.1

The variations of gravitational potential energy and kinetic energy of the car with time t are shown in Fig. 2.2.

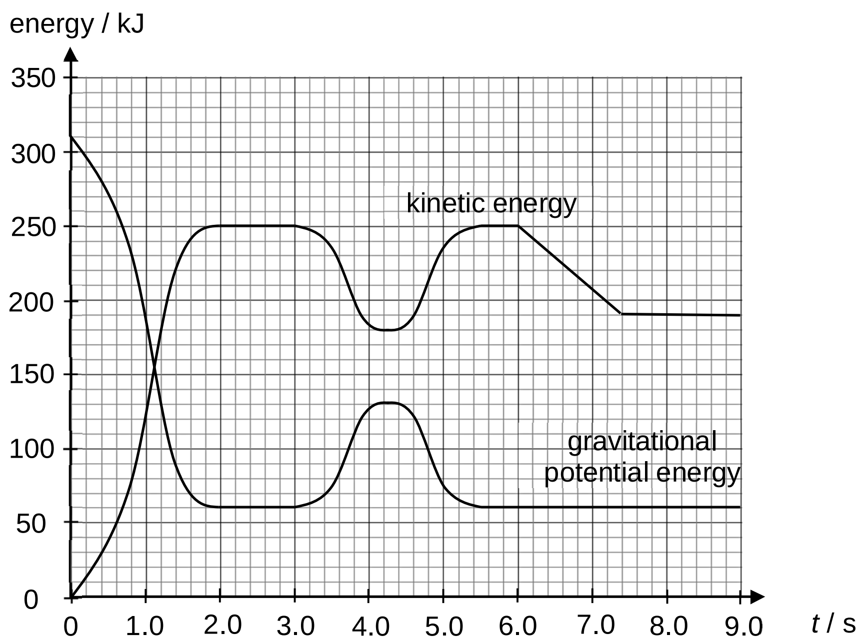

Fig. 2.2

(a) On Fig. 2.2, sketch the variation of total mechanical energy with time of the car, from t = 0 s to t = 9.0 s. [2]

(b) Between t = 3.0 s and t = 5.4 s, the car executes the loop.

Using Fig. 2.2, calculate d.

d = …………..……………..…..... m [2]

(c) (i) State the start and end times at which the car is within zone A.

start time = ……………..…………...…..... s

end time = ……………..…………..…..... s [1]

(ii) Calculate the rate at which the car loses kinetic energy in zone A.

rate = ……………..…………..…..... W [2]

(iii) Calculate the speed of the car when it first enters zone A.

speed = ……………..…………..…..... m s^{-1} [2]

(iv) Hence, calculate the magnitude of the frictional force acting on the car when it first enters zone A.

force = ……………..…………..…..... N [2]

3 (a) Explain what is meant by *internal energy* of a gas.

……………………………………………………………………………………………………

……………………………………………………………………………………………………

…………………………………………………………………………………………… [1]

(b) The pressure p of an ideal gas of density ρ is related to the mean-square speed of its molecules by the expression

$$p = \frac{1}{3}\rho\langle c^2\rangle$$

Show that the average kinetic energy of a molecule of an ideal gas is proportional to its thermodynamic temperature T.

[2]

(c) An engine operates by using 0.024 mol of an ideal gas. The gas undergoes a cycle of changes as shown in Fig. 3.1.

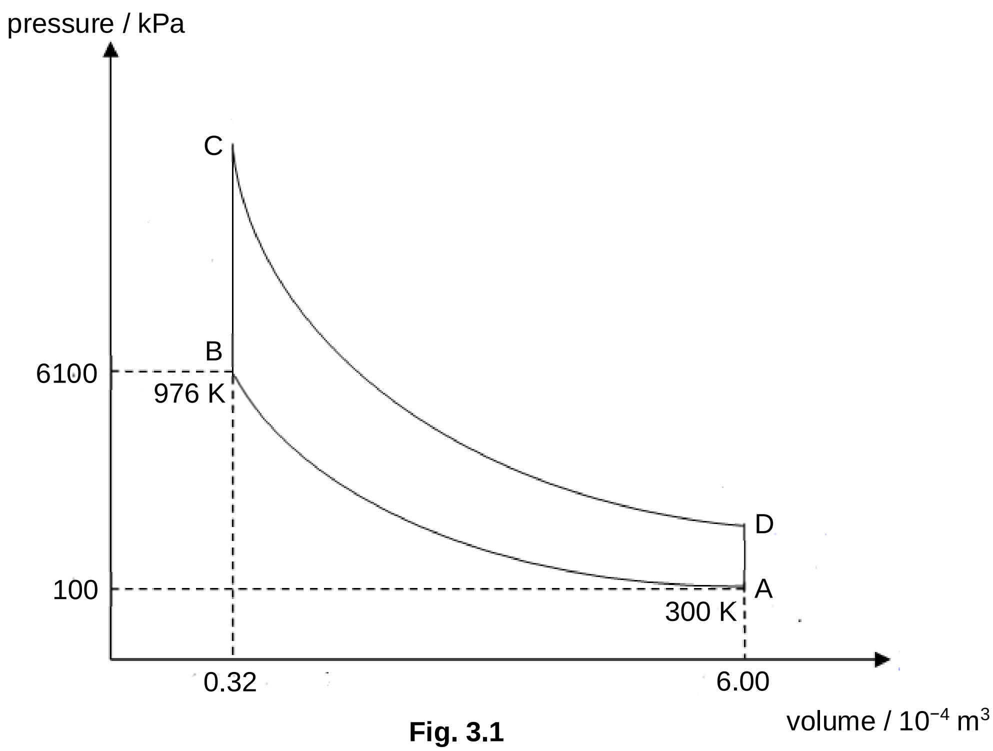

Fig. 3.1

The temperature of the gas at A and B are 300 K and 976 K respectively.

(i) Given that the gas gains 250 J of heat during a heating process at constant volume from B to C, determine

1. the temperature at C,

temperature = ... K [3]

2. the pressure at C.

pressure = ... Pa [1]

(ii) Fig. 3.2 shows some of the values of changes in Fig. 3.1.

Complete Fig. 3.2.

change	heat supplied to gas / J	work done by gas / J	increase in internal energy / J
A to B	0	–200	
B to C	250		
C to D	0		
D to A	–100		

Fig. 3.2

[3]

4 Fig. 4.1 shows two identical solenoids with their axes aligned and the two opposite faces at a distance of 0.20 m apart.

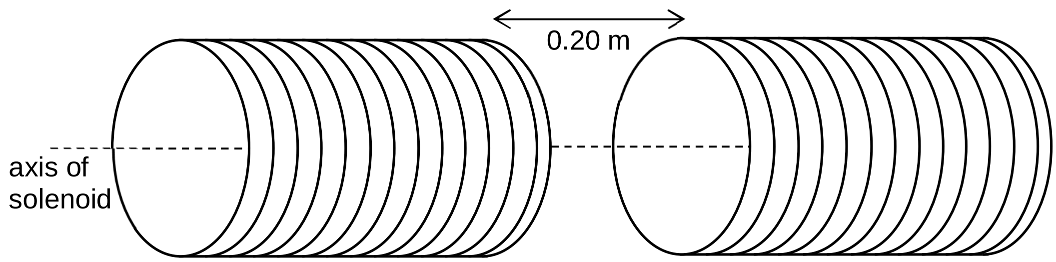

Fig. 4.1

The space between the solenoids is a vacuum with a uniform magnetic flux density of 1.2×10^{-3} T.

An electron moving with speed 5.0×10^{7} m s^{-1} normal to the magnetic field between the solenoids traces an arc of a circular path.

(a) (i) Determine the radius of the circular path.

radius = ……….………………...... m [2]

(ii) State and explain whether the electron will collide with any of the solenoids.

…………………………………………………………………………………………………

…………………………………………………………………………………………………

……………………………………………………………………………………………… [2]

(b) In another similar experiment described above, the electron is found to spiral inwards with decreasing radius even though the magnetic field remains constant.

With reference to the speed of the electron, suggest a reason for this motion.

....

....

.... [2]

(c) A uniform electric field is applied in the region of the magnetic field so that the electron continues undeflected through the two fields.

Determine the magnitude of the electric field strength *E*.

E = ... N C^{-1} [2]

5 **(a)** When monochromatic radiation is incident on a clean cadmium surface, electrons with a range of kinetic energies up to a maximum of 3.65×10^{-20} J are released. The work function of cadmium is 4.07 eV.

(i) Explain what is meant by *work function*.

……………………………………………………………………………………………

…………………………………………………………………………………………. [1]

(ii) Explain why the emitted electrons have a range of kinetic energies up to a maximum.

……………………………………………………………………………………………

……………………………………………………………………………………………

……………………………………………………………………………………………

…………………………………………………………………………………………. [2]

(iii) Calculate the wavelength of the radiation.

wavelength = ... m [2]

(b) In the wave model of electromagnetic radiation, electrons near the surface of a metal absorb energy from the wave.

(i) The radiation incident on the cadmium surface in **(a)** delivers energy at a rate of 3.0×10^{-22} J s^{-1}.

Calculate the minimum time it would take an electron to absorb sufficient amount of energy to be emitted from the metal surface.

time = ... s [1]

(ii) In the actual experiment conducted, photoelectrons are observed to be emitted without any time lag.

Explain how this observation provides evidence for the particulate nature of electromagnetic radiation.

....

....

....

.... [2]

6 Tritium is an isotope of hydrogen and is represented by the symbol ${}^{3}_{1}H$.

(a) Explain the term *isotope*.

....

.... [1]

(b) Tritium has a binding energy per nucleon of 2.83 MeV.

(i) Explain what is meant by the *binding energy* of a nucleus.

....

.... [1]

(ii) Given that the mass of a proton is 1.00783 u and the mass of a neutron is 1.00867 u, calculate, to five decimal places, the mass of a tritium nucleus.

mass = ... u [3]

(c) In a fusion power reactor, two deuterium nuclei fuse together to form a tritium nucleus.

(i) Complete the nuclear equation below for the fusion of deuterium.

$$ {}^{2}_{1}\mathrm{H} + {}^{2}_{1}\mathrm{H} \rightarrow {}^{3}_{1}\mathrm{H} + \ldots\ldots\ldots\ldots $$

[1]

(ii) For the fusion of deuterium shown in **(i)**, 4.03 MeV of energy is released.

Calculate

1. the binding energy of a deuterium nucleus,

binding energy = ... MeV [2]

2. the mass of deuterium required per unit time to produce 2.86×10^{9} W of power.

mass = ... kg [3]

7 The decay of radioactive materials is a random process. On average, nuclides which decay rapidly exist for a shorter time than nuclides which decay slowly. It is common practice when making calculations on decay to make use of the half-life of a nuclide. One difficulty that arises with these calculations is when the radioactive material is a mixture of two or more nuclides.

This question considers the case when a mixture of two radioactive nuclides is present. In decommissioning a nuclear power station, this difficulty is compounded by the presence of about a hundred different radioactive nuclides in significant quantities.

(a) Explain what is meant by a *random* process.

....

....

.... [2]

(b) Define *half-life*.

....

....

.... [1]

(c) $^{131}_{52}Te$ decays by β emission to $^{131}_{53}I$. $^{131}_{53}I$ is not stable, and decays by β emission to the stable isotope Xe, but the half-life for this decay is very much longer than that for the decay of $^{131}_{52}Te$. A sample of pure $^{131}_{52}Te$ is prepared at time t = 0, and Fig. 7.1 shows the variation with time of the activity A of the whole sample. Background radiation can be ignored.

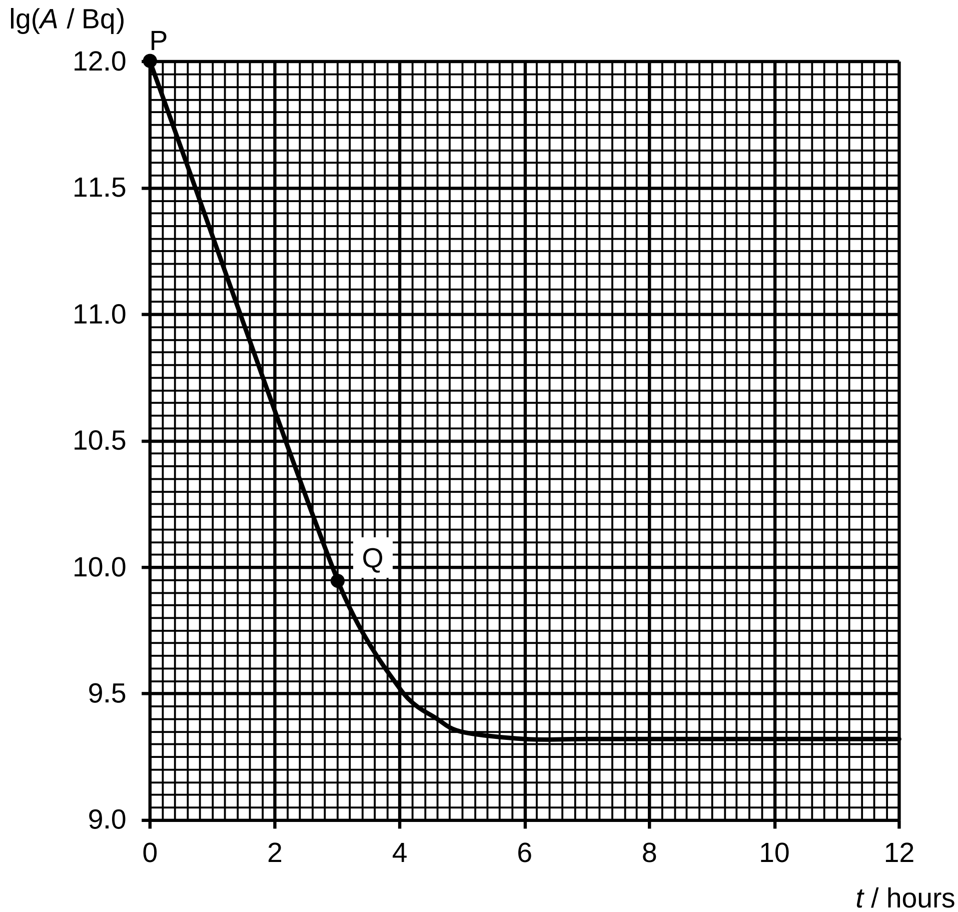

Fig. 7.1

(i) Write down the nuclear equation for the decay of $^{131}_{53}\mathrm{I}$ to Xe, showing the mass number and atomic number of the isotope Xe clearly.

.... [2]

(ii) Explain why the part PQ on the graph corresponds mainly to the decay of $^{131}_{52}\mathrm{Te}$.

....

....

.... [2]

(iii) Determine the gradient of PQ.

gradient = ... [2]

(iv) Hence, or otherwise,

1. show that the half-life of $^{131}_{52}\text{Te}$ is approximately 0.44 hours,

[3]

2. determine the initial number of $^{131}_{52}\text{Te}$ nuclides in the sample.

number of nuclides = ... [2]

(v) Explain why the graph seems flat after a time of 6 hours.

....

....

.... [2]

(vi) Hence, calculate the half-life of $^{131}_{53}\mathrm{I}$.

half-life = ... hours [3]

(d) Suggest why background radiation can be ignored in this question.

....

....

………………………………………………………………………………………… [1]

(e) State and explain whether these two radioactive nuclides would pose any hazard if found when de-commissioning a nuclear reactor.

....

....

………………………………………………………………………………………… [2]

End of paper